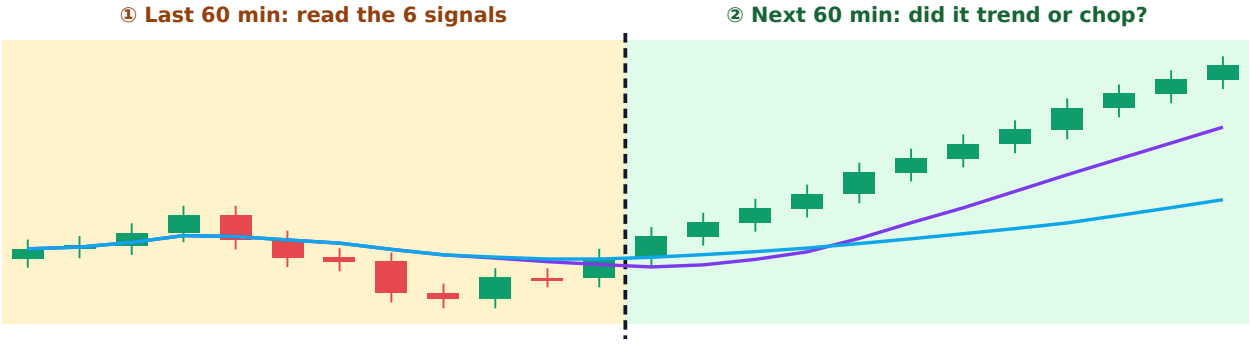
Sideways test: does "sideways" or "trending" hold?

PalTra · Scalp page → **Sideways test** tab. Read-only research on past Groww candles: no order is placed and nothing in the bots reads it. Run it on the backtest server.

Why this test

A stock that **was** sideways in the last hour is not always sideways in the **next** hour, and a "strong trend" at 10:00 often goes flat by 11:00. Routing stocks between bots, or switching a filter on, only helps if a reading **holds** long enough to trade. This test measures that on your own stocks and past days, instead of guessing.

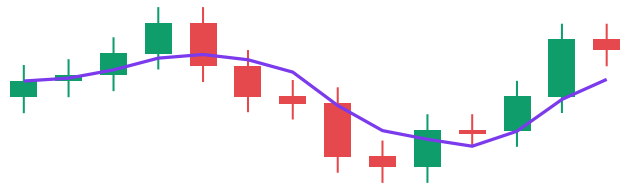


This happens at 10:00, 11:00, 12:00, 13:00, 14:00 and 14:30, every past day, every stock.

The six signals it reads (on the window just finished)

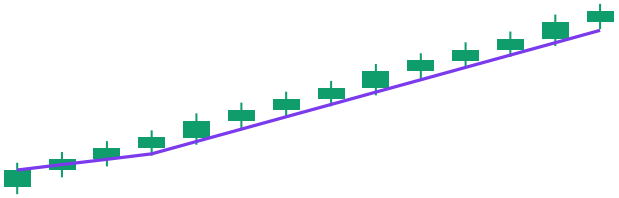
Signal	What it is	Sideways usually shows…
SMA crosses	How many times SMA 9 crossed SMA 21.	2 or more crosses in an hour
SMA 9 turns (V / U)	How many times the SMA 9 changed direction (tiny wiggles ignored).	2 or more turns
ADX 14	The trend-strength reading on the last closed candle.	below 20
Efficiency	Net move ÷ all the up-and-down moves. 1 = a straight line, 0 = back where it started.	below 0.3
Candle overlap	How much each candle covers the one before (0 = none, 1 = the same range).	above 0.6
SMA gap	How far apart SMA 9 and SMA 21 are, in % of the price.	under 0.05%

Sideways window



many crosses, low efficiency, big overlap

Trending window



no crosses, efficiency near 1, small overlap

"Trended" next window means the same rule the Trend quality phases use: it moved at least **1 ATR** and its efficiency was at least **0.5**. Everything else counts as sideways. The test also shows what simply following the SMA side would have made in that next window (enter the side the SMAs point to, flip on each cross, close at the end; before charges): a rough stand-in for the bot.

How to run it

1. Open the **backtest server** (it has its own Groww login, so it never slows the live bots), then **Scalp → Sideways test**.
2. **Scan:** pick the stocks: your armed stocks, Nifty 50, Nifty 100, Bank Nifty or all F&O (up to 200 are read).
3. **Past days:** 10 is a good start; 20-30 gives steadier numbers but takes longer.
4. **Candle (min):** the candle size your bot trades (e.g. 5 or 15).
5. **Window (min):** how long a reading must hold to be useful to you. 60 = "the next hour".
6. Press **Run test**. It reads one stock at a time ("Reading stock 12 of 50…"); you can leave the page. **Stop** keeps the stocks read so far.

> On the main desk it is refused during market hours unless the desk is set to run backtests at any time, because it shares the bots' Groww limit there. The last result is kept, so you can come back to it.

How to read a signal card (example numbers)

Example: **SMA crosses in the last window**, base rate 38% (on all checks, the next hour trended 38% of the time).

Reading	Checks	Next window trended	Next crosses	SMA-follow P&L
0	420	51% (+13)	0.6	+0.041%
1	510	40% (+2)	1.1	+0.006%
2	300	29% (-9)	1.7	−0.032%
3+	190	22% (-16)	2.4	−0.071%

Column	Meaning
Reading	The signal's value at the check time (here: 0, 1, 2 or 3+ crosses in the last hour).
Checks	How many (stock, day, time) checks fell in that bucket. Grey rows have under 30 checks: too few to trust.
Next window trended	How often the next window trended, with the difference from the base rate. The black tick on the bar is the base rate.
Next crosses	Average SMA crosses in the next window: the whipsaws you would have met.
SMA-follow P&L	Average result of following the SMA side in the next window, % of price, before charges.

> **Reading this example:** after 0 crosses the next hour trended 51% of the time (+13 above the base rate); after 3+ crosses only 22% (−16), with 2.4 more whipsaws and a loss. The buckets are **29 points apart** (the **spread**), so this signal really separates trending from sideways hours on these stocks.

Spread	Verdict on the card
20 points or more	**strong**: the buckets lead to very different next hours. Worth using.
10 to 20 points	**some**: worth testing as a filter in a backtest.
under 10 points	**weak**: about the same as guessing. Don't build on it.

What to do with the result

The cards are sorted with the strongest signal first, and the line above them names the best one. Turn it into a setting, then **prove it with a replay** before using it live:

If this signal is strong…	Try this in Settings (SMA strategy)
SMA crosses	6. Size, daily limits → **Max trades per stock** = 2, so a stock that keeps crossing stops being traded. Or a bigger candle.
SMA gap	2. Checks before entering → **SMA gap range** (skip crosses with a tiny gap), or 4. **SMA gap mode** (enter only once the gap opens).
ADX	2. Checks before entering → **ADX check** at the level whose bucket trended best (e.g. 25).
Efficiency / overlap	No setting reads these yet. Tell Claude which level worked; it can be built as an entry check or as the Bot 1 → Bot 2 / Bot 3 sorter.
SMA 9 turns	3. When to exit → the **V-turn** numbers, or 1. When to enter → the **SMA 9 turn entry** with "only toward the cross".

Then check it: save the setting on a PAPER bot, open the replay bar, pick the same past days and press Play. Compare the run's P&L, win rate and number of trades with a run without the change (Backtests tab). Keep a change only if it wins on most days, not just one.

If no signal is strong

Then none of these readings can tell you early enough which stocks will stay sideways. Moving stocks between bots on them would only repeat the same mistakes automatically. The better answer is to **trade less in chop**: a per-stock trade cap of 2, a bigger candle, gap mode, and a daily loss limit, and to keep sizes small.

Good to know

- The test looks only at the past window for its readings and only at the next window for the result: there is no peeking ahead.
- Repeat it every few weeks: markets change between trending and range-bound months.
- The SMA-follow P&L is before charges and is not your bot's exact settings; use a replay for the real number.
- Different stocks behave differently. Run it on the list you actually trade.